

SHREYAS KARNAD

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SUMMARY

Health Informatics graduate (University of Michigan) and Biomedical Engineer with experience in biomedical research, health-care data analysis, and computational research workflows. Work includes experimental sensor analysis, clinical requirements gathering, SQL-based data review, and AI-assisted analytics. Co-author of a peer-reviewed review on AI in digital pathology, with experience communicating findings to clinical, research, and business stakeholders.

EDUCATION

University of Michigan – School of Information, Ann Arbor, MI Aug 2024 – Apr 2026

Master of Health Informatics (Machine Learning, Data Analytics) — GPA: 3.936

Focus: health data analytics, clinical decision support, FHIR interoperability, AI/ML in healthcare, data governance

Manipal Institute of Technology, MAHE, Manipal, India Nov 2021 – Aug 2023

M.Tech, Biomedical Engineering (Biomedical Instrumentation) — GPA: 3.86/4.00 (WES-evaluated)

MIT School of Bioengineering Sciences & Research, MIT ADT, Pune, India Aug 2017 – Aug 2021

B.Tech, Bioengineering — GPA: 3.93/4.00 (WES-evaluated)

RESEARCH TRAINING

CITI Program – Human Research: Data or Specimens Only Research Jun 2026

Completed Basic Course covering research ethics, IRB regulations, records-based research, and HIPAA privacy protections. Expires Jun 2029.

EXPERIENCE

Northwell Health Equity Partners, New York, NY Jul 2026 – Sep 2026

Administrative Intern

- Designed and implemented, with AI assistance, a relational database schema (SQLite) extending NHEP's production AI research agent, including company records, aliases, and an append-only membership-history table; shipped as seven independently reviewed pull requests, growing the test suite from 265 to 457 passing tests.
- Built automatic entity matching connecting news items to tracked companies by name and alias, with whole-word matching to avoid false positives and ambiguous matches left unassigned rather than guessed; the final release (v0.8.0) was reviewed, merged, and deployed to production, confirmed working on live scheduled briefs.
- Built four query tools (company directory, single-company lookup, full history, and unmatched-news search) giving the investment team direct access to company and news data through the team's existing chat-based interface.
- Filtered a 12,000+ entry contact database to 305 qualified prospects against supplied investor criteria, cross-referencing outreach history and investor type, with SQL queries to verify firms; leadership approved the methodology and used the shortlist directly.
- Built Excel-based dashboards tracking the deal pipeline, giving the investment team visibility into deal stage, sourcing, and follow-up status across the portfolio.

University of Michigan – School of Public Health, Ann Arbor, MI Jan 2026 – May 2026

Grader, Public Health Informatics (Part-time)

- Evaluated 98+ graduate-level assignments per cycle using structured rubrics, maintaining consistent scoring standards and providing detailed written feedback on analytical approach and data interpretation.
- Collaborated with Prof. Rahul Ladhanian on grading rubric design and quality assurance across multiple assignment cycles; identified and corrected systematic grading inconsistencies before grade release.

University of Michigan – School of Information, Ann Arbor, MI Sep 2025 – Dec 2025

Instructional Aid, Health Informatics / Database Management (Part-time)

- Supported a 150+ student course by grading analytical examinations, proctoring assessments, and communicating technical concepts clearly to students with varying programming backgrounds.
- Collaborated with Prof. Bobby Madamanchi on exam paper design and quality assurance across 3 exam cycles; identified and corrected systematic grading inconsistencies before grade release.

Biocomplexity Institute, Indiana University, Bloomington, IN (Remote) Jun 2025 – Aug 2025

Data Science Intern

- Contributed to a workflow translating natural-language biological research descriptions into structured parameters and CompuCell3D simulation code, with guidance from Indiana University researchers.
- Evaluated generated outputs using AI-recommended tests and rating criteria; documented version mismatches, hallucinated API calls, and incorrect biological parameters to support refinement.

- Helped adapt a nine-agent CrewAI workflow to DeepSeek through Replicate and a custom Python wrapper, resolving integration and environment conflicts.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India (Remote)

Aug 2023 – Aug 2024

Associate Biomedical Engineer

- Evaluated deep learning classification models (ResNet50, GoogLeNet, DenseNet) on magnetic-probe signal data for sentinel lymph node detection research, and wrote up comparison results for clinical stakeholders in plain terms.
- Tested an early-stage remote-processing prototype using EC2-hosted MATLAB classification code and S3 storage, using basic geometric inputs to check functionality; the intended application was to offload computation from hospital computers.
- Integrated MATLAB software with LabVIEW-based hardware interfaces to route sensor data from the magnetic field probe into the classification pipeline.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India

Aug 2022 – Jul 2023

Graduate Research Intern

- Interviewed 20+ surgical oncologists to translate clinical workflow needs into formal technical requirements for the engineering team.
- Built and tested a magnetic probe prototype for intraoperative sentinel lymph node detection; analyzed experimental sensor data using time-series methods and contributed findings to my M.Tech thesis and internal technical reports.

TATA Elxsi, St. Paul, MN (Remote)

Jan 2021 – Jun 2021

Project Intern

- Analyzed investment data across 50+ digital health companies totaling over \$12B and prepared sector rankings and a recommendation memo used for leadership planning discussions.
- Reviewed FDA guidance and IMDRF frameworks for Software as a Medical Device (SaMD), preparing a reference guide for product development planning.

ACADEMIC AND INDEPENDENT PROJECTS

MIMIC-IV Health Equity & Disparities Dashboard – AI-Assisted Learning Project

Jun 2026

- Built a six-page Power BI report with AI guidance using BigQuery staging and aggregation queries covering 546,028 hospitalizations; examined query logic and configured the report's data model and visuals.
- Investigated an incorrect 289% discharge-rate value with AI guidance; corrected visual field assignments and applied revised DAX calculations using admission counts instead of summed subgroup percentages.

MIMIC-IV Sepsis Modeling Exploration – AI-Assisted Learning Project

Jun 2026

- Pursued a sepsis-modeling exploration inspired by Michigan Medicine research; executed and examined AI-generated BigQuery SQL and Python for a cohort of 67,286 ICU stays.
- Re-exported the feature dataset after the notebook loaded only 29,405 rows, then checked that the loaded dataset matched the 67,286-row BigQuery cohort.
- Ran Random Forest training and explored feature attribution and FHIR-formatted prediction outputs using AI-generated code. GitHub: github.com/karnadsp/mimic-sepsis-ews

APP-2-MARKET Product Launch Strategy (Academic Simulation) – SI 661, University of Michigan

Jan 2026 – Apr 2026

- Led project planning for a simulated health data app's go-to-market strategy, structuring the work into a 6-phase, 30+ task Work Breakdown Structure for a hypothetical company (Tradarbol Tech).
- Built a scenario project budget (\$92,000 projected vs. \$93,150 simulated actual costs) and defended sequencing trade-offs to classmates under live Q&A.

EHR AI Hallucination Mitigation Framework (Research Proposal)

Mar 2026 – Apr 2026

- Identified that AI-generated hallucinations in EHR transcription tools can create undetected clinical documentation errors, and proposed a Zero-Trust RAG framework with retrieval-based validation checkpoints intended to reduce reliance on unverified LLM outputs.

Ctrl Alt Drink – Alcohol Moderation App Concept (Team Project) – SI 554, UMich

Jan 2026 – Apr 2026

- Contributed to a 3-person team project designing a conceptual mobile app for student alcohol moderation, based on published research on unhealthy alcohol use; wrote the intended users, user needs, and technological platform sections of the team paper.
- Designed Figma wireframes for a clinician-contact feature flow and contributed to the final presentation.

Structural Health Disparities and Sepsis Risk Analysis (Team Project) – SI 624, University of Michigan

Sept 2025 – Dec 2025

- Contributed the Random Forest classification component (80% accuracy) and feature-importance visualizations to a team study of county-level sepsis outcomes, and helped integrate the findings into the academic report.

Breast Cancer Tumor Network Analysis – SI 507, University of Michigan

Feb 2025 – Apr 2025

- Compared K-Means and Spectral Clustering on a 569-record breast-tumor diagnostic dataset in Python; used three network-

centrality measures to examine relationships among tumor features.

Omega Hospital Service Line Evaluation – HMP669, University of Michigan

Mar 2025 – Apr 2025

- Cleaned and standardized hospital records in Microsoft Access; worked with a three-person team in RStudio to analyze cardiology and orthopedic service-line finances and develop recommendations for an academic report.

Diabetes Risk Assessment App + FHIR Output (Team Project)

Oct 2024 – Nov 2024

- Contributed to a team-developed diabetes risk application that included an ML pipeline and FHIR-formatted outputs to explore clinical decision support and interoperability concepts.

Diabetes Dataset Analysis – SI 544, University of Michigan

Sep 2024 – Dec 2024

- Analyzed 768 records in R using two-sample t-tests and multiple linear regression of glucose and insulin levels; flagged zero-valued clinical measurements as potential missing data.

IVF Success Prediction – Manipal Institute of Technology

Apr 2022 – Jun 2022

- Compared Logistic Regression, K-Nearest Neighbour, and Naïve Bayes classifiers in Python using approximately 55 patient records containing clinical parameters and 14 metabolite measurements.
- Evaluated six metabolite combinations; the best reported KNN result classified 10 of 12 evaluation cases correctly (83%), with limited sample size identified as a constraint.

PPI2 Web Server – Protein-Protein Interaction Database

Jan 2019 – Dec 2020

- Built the HTML-based user interface and connected it to a relational biological database via a JDBC driver, supporting researcher queries and data exploration.

PUBLICATION

Co-author, *Current trends of artificial intelligence and applications in digital pathology: A comprehensive review*. International Journal of Advanced and Applied Sciences, 10(12), 29–41, Dec 2023.

SKILLS

Applied Experience: SQL, Python, R, MATLAB, LabVIEW, Microsoft Access, Power BI (learning projects), Excel, Word, PowerPoint, Git/GitHub | **Statistical/ML Methods:** hypothesis testing, regression, classification, time-series analysis | **Project Exposure:** BigQuery, SQLite, AWS EC2/S3 (prototype testing), FHIR R4 output generation, AI-assisted workflows (CrewAI, DeepSeek, prompt engineering) | **Healthcare/Research Training:** FHIR project exposure; HL7 and HIE coursework; CITI human research training covering IRB regulations and HIPAA privacy protections